

Experiments: Part II

PS200C Quant Methods in Politics, Spring 2026

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Where we are

- Part I: random assignment makes the simple difference in means an unbiased estimator of the ATE.
- Part II: **unbiased is not enough**. The same experiment can be run well or badly. Today is about what “well” means and how design and analysis together deliver it.
- Roadmap:
 - 1 Why unbiasedness isn't the whole story (and balance as a hook)
 - 2 Covariate adjustment (precision, not bias)
 - 3 Blocking (better-than-random balance by design)
 - 4 Non-independence: dependent residuals and clustered assignment
 - 5 Wrap and takeaways

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- For an unbiased estimator, $\text{RMSE} = \text{SD}$. **Reducing variance is the entire game.**

Two different “variances”

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- These can be very different things. A bad SE estimator can produce tight intervals around a noisy estimate (under-coverage), or vice versa.
- Today: **HC2 (heteroskedasticity-robust)** SEs from `lm` via the `sandwich` package are the practical default for randomized experiments. We’ll see they work well for individual assignment — and badly for clustered assignment.

Running example: a GOTV experiment

- **Unit:** a registered voter.
- **Treatment D :** a get-out-the-vote contact (mailer, canvass, etc.) vs. no contact.
- **Outcome Y :** did they vote? (0/1)
- **Covariates X :**
 - `past_turnout`: an index of voting in the last several elections. Strong predictor of Y . (This is a pre-treatment proxy for $Y(0)$.)
 - `age`, `partisan strength`: weaker predictors.
- **Structure:** voters are nested in households (4 per household) and households in precincts (10 per precinct). 100 precincts, $N = 4000$.
- **True effect:** $\tau = +0.05$ on the probability of voting.
- **We will estimate $\hat{\tau}$ many times under different designs and analyses**, and compare the resulting distributions.

What does a single experiment look like? Check balance.

We randomly assign half the sample to treatment. Then look at the covariate means by treatment status:

Variable	Treated	Control	Difference
past_turnout	-0.028	-0.001	-0.027
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- Why we “check balance” in experiments isn’t to detect bias — randomization handles bias. It’s to see what kind of *chance variation* this realization generated, since that variation is what inflates $\text{Var}(\hat{\tau})$.
- Imbalance on a strong predictor of $Y \Rightarrow$ noisier $\hat{\tau}$. Imbalance on an irrelevant variable $\Rightarrow$ doesn’t matter.

Aside: Y is binary. Can we use diff-in-means / OLS?

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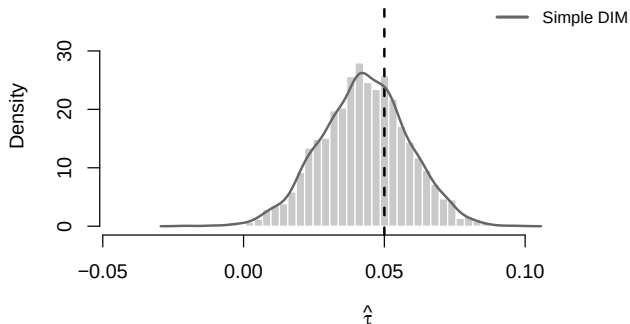
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- Logit/probit are fine too — but their coefficients are *not* the ATE. You'd have to compute marginal effects to recover the ATE.
- Punch line: don't let the binary outcome scare you off OLS. We'll use it throughout the simulations today.

Simulating the simple design

2,000 replications. Each one: assign D , draw Y , run $\text{lm}(Y \sim D)$ and grab the HC2 SE from `sandwich`.

Simple randomization, diff-in-means



Actual SD of $\hat{\tau}$ across reps: 0.016. **Mean HC2 SE** from a single `lm`: 0.016. The estimator is wide, but the SE accurately knows it.

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- Rule of thumb: *covariate adjustment helps to the extent X predicts Y , not to the extent X “is a variable you have.”*
- Strong predictors (`past_turnout`) help a lot. Weak predictors (`partisan_strength`) barely help. There's no harm in including both, in moderation.

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- So whatever variance was being driven by that imbalance also goes away.
- (Formal version — the Frisch-Waugh-Lovell theorem — in the appendix.)

The Lin estimator (one slide)

The simplest covariate-adjusted estimator is just:

$$Y_i = \alpha + \tau D_i + \beta' X_i + \varepsilon_i \quad (\text{plain OLS, fine in large samples})$$

Lin (2013) showed a small modification is essentially always weakly better:

$$Y_i = \alpha + \tau_{Lin} D_i + \beta'_1 (X_i - \bar{X}) + \beta'_2 D_i \cdot (X_i - \bar{X}) + \varepsilon_i$$

- Demean X , then interact with D .
- τ_{Lin} is interpretable directly as an ATE.
- Removes a small ($O(1/N)$) bias from plain regression adjustment that Freedman (2008) flagged.
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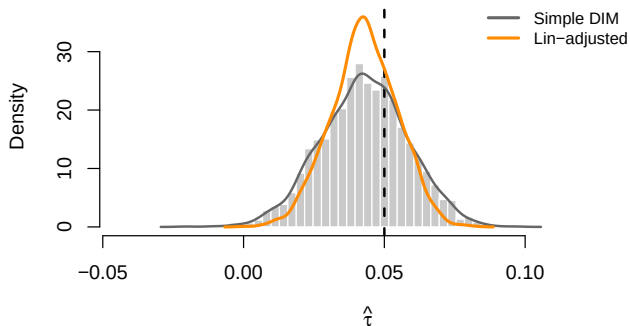
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(Detailed derivation, why demeaning matters, Freedman vs. Lin discussion: appendix.)

Simulation: + Lin-adjusted

Add regression adjustment (Lin)



Same 2,000 experiments. The grey bars are the simple-DIM baseline. The orange curve is the Lin-adjusted estimator. Same center, $\sim 25\%$ tighter.

SD(actual) = 0.012 Mean HC2 SE = 0.012 ($\Rightarrow$ HC2 still nails it)

Blocking: build the balance in, don't hope for it

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- **Slogan:** block what you can, adjust for what you can't.

Blocking: the estimator

Within each block j , compute the block-specific diff-in-means $\hat{\tau}_j$. Then weight by block size:

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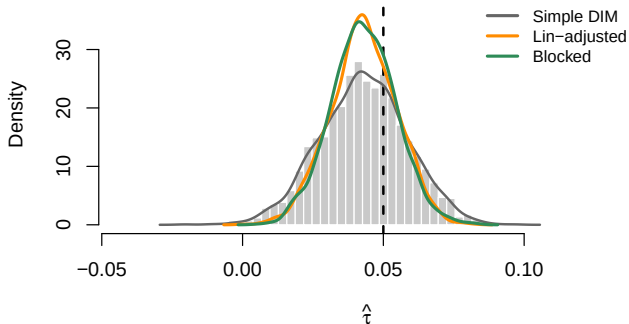
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- Caveat: with unequal treatment probabilities across blocks, plain block-FE OLS gives a (weighted) ATE that may not be what you want; use the block-weighted form above. (Aside on “weird weighting” later in the course.)

Simulation: + blocked

Add blocking on past-turnout quartiles



Add the green curve: blocking on `past_turnout` quartiles. Lands essentially on top of the Lin-adjusted curve in this DGP.

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Sometimes blocking strictly beats post-hoc adjustment; in well-behaved cases the two converge.

When does blocking help most? Intuition

- Blocking helps a lot when the blocking variable *strongly predicts* Y . Then within-block variation in Y is small, so the diff-in-means within each block is sharp.
- Blocking on something unrelated to Y is essentially free, but also essentially useless.
- In our sim, we blocked on `past_turnout`, which is the dominant predictor of voting — so we got a large gain.
- (Formal variance comparison in the appendix.)

A canonical real example: Gerber, Green, & Larimer (2008)

- Field experiment on social pressure and voter turnout (Michigan primary, 2006).
- Several mailings, including one revealing each voter's past turnout to their neighbors.
- **Design:**
 - Blocked on household composition and prior vote history.
 - Randomized at the *household* level, not the individual level. (Foresadows the next section.)
- **Result:** the “neighbors” treatment moved turnout by ≈ 8 percentage points — enormous for a piece of mail.
- **Why we cite this design:** the blocking + household clustering match the situation we're simulating today. Their estimator is precise because the design earned it, not because the analysis fixed it.

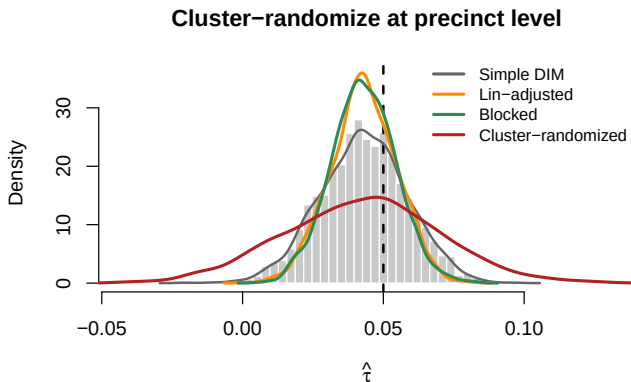
Two problems from non-independence

So far we've assumed observations are (conditionally) independent. Real settings often violate this — voters in the same precinct, students in the same classroom, repeated measures on the same person.

Two distinct things can go wrong, and they need different fixes:

- 1 **Clustered assignment.** Treatment is assigned at the group level (precinct, village, school). The point estimate itself becomes noisier — the experiment is worse.
- 2 **Dependent residuals.** Even with individual assignment, residuals can be correlated within a group, and SEs that ignore this can be wrong.

Simulation: cluster-randomize at the precinct level



Same DGP, same τ , but treatment is now assigned at the **precinct** level (everyone in a precinct gets the same D). The red curve is enormously wider than the individual-assignment baseline. Each precinct contributes essentially one “decision” — with 100 precincts, that’s a much smaller experiment than $N = 4,000$ suggests.

HC2 is wrong here. CR-SE fixes it.

Estimator	SD(actual)	HC2 SE	CR-SE (precinct)
Simple DIM ($\text{lm}(Y \sim D)$)	0.016	0.016	0.016
Lin-adjusted	0.012	0.012	—
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- HC2 assumes independent observations. With cluster-level assignment everyone in a precinct shares the same D — that assumption is badly violated.
- Cluster-robust SE (clustered at the precinct level) recovers the actual sampling variance almost exactly. In R:

```
sandwich::vcovCL(fit, cluster = precinct_id).
```
- The CR-SE is not magic — it estimates the actual within-cluster residual covariance from the data, not the worst-case “ N = number of clusters.”

When do you actually need CR-SEs? The principle

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Either ingredient *alone* is harmless:

- No cluster-shared component $\Rightarrow$ residuals are iid $\Rightarrow$ HC2 is fine. (Even cluster-randomized assignment has no penalty — we verified this in simulation.)
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Both at once is the dangerous case. Cluster-shared u_c *and* no within-cluster treatment variation to break it up. That is exactly our cluster-randomized precinct example.

Where “both at once” shows up in practice

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Subtle but real: “students within schools” with individual, balanced random assignment within school is *not* automatically a case for CR-SEs. It usually is in practice — because one of (1)–(5) is hiding in the design — but not from

Catches with CR-SEs

- **They are not free.** CR-SEs can be substantially conservative when within-cluster residual correlation is weak — intervals widen, power drops, with no coverage gain.

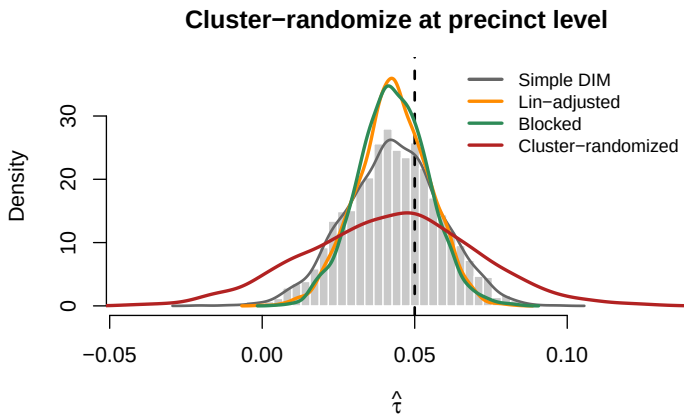
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- **Fixes for few clusters:**
 - Wild cluster bootstrap (Cameron, Gelbach & Miller 2008):
`fwildclusterboot` in R, `boottest` in Stata.
 - Randomization inference at the cluster level.
 - Aggregate to the cluster level and run the analysis there.

Same experiment, four designs and analyses



Same DGP, same true $\tau = 0.05$. None of these estimators is biased — they differ entirely in precision. Both *design* (blocking, avoiding unnecessary clustering) and *analysis* (covariate adjustment) deliver real gains.

Takeaways

- **Unbiasedness is not enough.** The same experiment can be run more or less precisely; that's what we're paying attention to today.
- **Adjustment works.** Covariate adjustment recovers real precision when X predicts Y . Use Lin's interacted form, with HC2 SEs, and pre-specify the covariates.
- **Design works too, often without paying anything for it.** Block on what you can; adjust for what you can't. The two are complements, not substitutes.
- **Cluster on the source of dependence in your residuals**, not on every group you can think of. Watch the few-clusters case and reach for the wild bootstrap when needed.
- Next: what if we can't randomize at all? **Selection on observables.**

Appendix

Frisch-Waugh-Lovell (FWL) theorem

Two facts:

- 1 Any bivariate regression of Y on X has $\hat{\beta} = \widehat{\text{Cov}}(Y, X) / \widehat{\text{Var}}(X)$.
- 2 The coefficient on the k th regressor in a multivariate regression can be obtained by (i) regressing X_k on the other X 's and saving the residual $\tilde{X}_k$, then (ii) regressing Y (or its residual $\tilde{Y}$) on $\tilde{X}_k$.

Together:

$$\hat{\beta}_k = \frac{\widehat{\text{Cov}}(\tilde{Y}_i, \tilde{X}_{ki})}{\widehat{\text{Var}}(\tilde{X}_{ki})}$$

Why experimental findings are robust to specification

Apply FWL to D :

$$\hat{\beta}_D = \frac{\widehat{\text{Cov}}(\tilde{Y}_i, \tilde{D}_i)}{\widehat{\text{Var}}(\tilde{D}_i)}$$

where $\tilde{D}_i$ is the residual from regressing D_i on X_i .

- Under randomization, $D \perp\!\!\!\perp X$, so on average $\tilde{D} \approx D$.
- Hence adjusting for X doesn't move $\hat{\beta}_D$ much in expectation — it just removes the variance contribution from chance imbalances.
- This is why experimental results are typically robust to “what did you control for?”

Lin (2013): why the demeaning?

$$Y_i = \alpha + \tau_{Lin} D_i + \beta'_1 (X_i - \bar{X}) + \beta'_2 D_i (X_i - \bar{X}) + \varepsilon_i$$

- $\mathbb{E}[Y_i \mid D_i = 0, X_i = \bar{X}] = \alpha$
- $\mathbb{E}[Y_i \mid D_i = 1, X_i = \bar{X}] = \alpha + \tau_{Lin}$
- So τ_{Lin} is the comparison of treated vs. control *at the mean of X* .
- Because $\mathbb{E}_X[\text{ATE}(X)] = \tau_{Lin}$ for this model, it recovers the ATE without assuming constant treatment effects.
- Plain OLS without the interaction works in large samples but is biased $O(1/N)$ (Freedman 2008). Lin's form removes this and is now standard practice.

When does blocking help? (variance comparison)

Models for complete and block-randomized designs:

$$Y_i = \alpha + \tau_{CR} D_i + \epsilon_i$$

$$Y_i = \alpha + \tau_{BR} D_i + \sum_{j=2}^J \beta_j B_{j[i]} + \epsilon_i^*$$

Then

$$\text{Var}[\hat{\tau}_{CR}] = \frac{\sigma_\epsilon^2}{\sum_i (D_i - \bar{D})^2}$$

$$\text{Var}[\hat{\tau}_{BR}] = \frac{\sigma_{\epsilon^*}^2}{\sum_i (D_i - \bar{D})^2 (1 - R_D^2)}$$

where R_D^2 is from regressing D on the block dummies.

When does blocking help? (intuition)

- If treatment proportions are the same in every block, $R_D^2 = 0$ and the denominator is unchanged — blocking doesn't *cost* you anything.
- If the blocks strongly explain Y , then $\sigma_{\varepsilon^*}^2 \ll \sigma_{\varepsilon}^2$ — blocking *wins* a lot.
- Combine: blocking on a strong predictor of Y , with equal-share randomization, weakly improves precision and usually does so a lot.

Cluster-robust variance: a sketch

For OLS $\hat{\beta} = (X'X)^{-1}X'Y$, the cluster-robust variance is

$$\widehat{\text{Var}}_{CR}(\hat{\beta}) = (X'X)^{-1} \left(\sum_{c=1}^C X'_c \hat{e}_c \hat{e}'_c X_c \right) (X'X)^{-1}$$

- X_c and $\hat{e}_c$ are the design matrix and residuals for cluster c .
- This collapses cluster-level scores, so it allows arbitrary intra-cluster correlation in residuals while assuming clusters are independent.
- The “effective $N = C$ ” approximation is the limit case $\rho \rightarrow 1$. The actual penalty is data-driven via $\hat{e}_c$.
- With $C \lesssim 30$, this estimator is downward biased; use a wild cluster bootstrap (Cameron, Gelbach, & Miller 2008) instead.

LPM caveats (binary outcomes)

Using OLS / LPM with a binary Y :

- **Pro:** unbiased for the ATE under random assignment; coefficients are interpretable as probability differences; works in every standard software environment.
- **Cons:**
 - Predicted probabilities can fall outside $[0, 1]$.
 - Heteroskedasticity is built in — use robust SEs.
 - In observational settings with extreme covariates, linearity can be a poor approximation.
- For the experimental settings in this lecture, none of these are reasons to abandon LPM.